

Matrices and Systems of Linear Equations (2022–2026)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

36

TOTAL MARKS

62 M

TEST DURATION

108 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	22	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	3	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	11	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q59 • ID: JAM_2026_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59

Let A be a 3×3 real matrix such that given any column vector $x \in \mathbb{R}^3$, the column vector Ax is the reflection of x about the plane

$$\{(a, b, -a - b) : a, b \in \mathbb{R}\}.$$

Then the sum of the diagonal elements of A is _____ (rounded off to one decimal place).

Question 2 (JAM 2026 • Q57 • ID: JAM_2026_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Let $a, b \in \mathbb{R}$. If the system of linear equations

$$x + 2y + 2z = 1$$

$$2x + 3y + z = 2$$

$$ax + 5y + bz = b$$

has infinitely many solutions, then $a + b =$ _____ (rounded off to one decimal place).

Question 3 (JAM 2026 • Q56 • ID: JAM_2026_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56

Let $M \in M_3(\mathbb{R})$. If

$$M \begin{pmatrix} \sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ \sqrt{2} \\ -5 \end{pmatrix}, \quad M \begin{pmatrix} 0 \\ 0 \\ \sqrt{30} \end{pmatrix} = \begin{pmatrix} \sqrt{15} \\ \sqrt{10} \\ \sqrt{5} \end{pmatrix}, \quad M \begin{pmatrix} -\sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{12} \\ -\sqrt{18} \\ 0 \end{pmatrix}$$

then $\det(M) =$ _____ (rounded off to one decimal place).

Question 4 (JAM 2026 • Q50 • ID: JAM_2026_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50

Let P be a 5×5 real matrix with $\det(P) = 2$. Let Q be the matrix of cofactors of P . Then $\det(Q) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 5 (JAM 2026 • Q49 • ID: JAM_2026_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49

Let $A \in M_3(\mathbb{C})$. Suppose the column vector $v = \begin{pmatrix} \sqrt{5}i \\ 2i \\ x \end{pmatrix}$ in $\mathbb{C}^3$ belongs to the intersection of $\text{nullspace}(A)$ and $\text{rangespace}(A^T)$.
Then $|x| = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 6 (JAM 2026 • Q39 • ID: JAM_2026_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39	Consider matrices P of order 4×6 and Q of order 6×4 with real entries such that $PQ = 0$ and $QP = 0$. Which of the following statements is/are TRUE?
(A)	$\text{rangespace}(P) \subseteq \text{nullspace}(Q)$ and $\text{rangespace}(Q) \subseteq \text{nullspace}(P)$.
(B)	$\text{rank}(P) + \text{rank}(Q) \leq 4$.
(C)	If $\text{rangespace}(P) = \text{nullspace}(Q)$, then $\text{rank}(P) + \text{rank}(Q) = 4$.
(D)	$\text{rangespace}(Q) = \text{nullspace}(P)$.

Question 7 (JAM 2026 • Q37 • ID: JAM_2026_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37

$$\text{Let } P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \in M_5(\mathbb{C}).$$

Which of the following statements is/are TRUE?

- (A) $\text{nullity}(P - I) \geq 2$, where I is the 5×5 identity matrix.
- (B) P has 4 distinct eigenvalues in $\mathbb{C}$.
- (C) P has 3 distinct eigenvalues in $\mathbb{R}$.
- (D) If λ is an eigenvalue of P , then there exists a positive integer n such that $\lambda^n = 1$.

Question 8 (JAM 2026 • Q25 • ID: JAM_2026_Q25)

MCQ

+2 M

Neg: -0.67

Q.25	Let P be a nonzero complex $n \times n$ matrix such that $v^* P v \geq 0$ for all column vectors v in $\mathbb{C}^n$. Which ONE of the following statements is necessarily TRUE?
(A)	All eigenvalues of P are negative real numbers.
(B)	If v_1, v_2 are column vectors in $\mathbb{C}^n$ such that $P v_1 = v_2$ and $P v_2 = v_1$, then $v_1 = v_2$.
(C)	$v^* P v \neq 0$ for all nonzero column vectors v in $\mathbb{C}^n$.
(D)	All eigenvalues of P^* are negative real numbers.

Question 9 (JAM 2026 • Q14 • ID: JAM_2026_Q14)

MCQ

+2 M

Neg: -0.67

Q.14	Let A be a 5×3 real matrix of rank 2 and b be a non-zero 5×1 real column vector. Suppose $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ are two solutions of the system of linear equations $Ax = b$. Which ONE of the following is also a solution of $Ax = b$?
(A)	$\begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$
(B)	$\begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix}$
(C)	$\begin{pmatrix} 5 \\ 6 \\ 9 \end{pmatrix}$
(D)	$\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$

Question 10 (JAM 2026 • Q8 • ID: JAM_2026_Q8)

MCQ

+1 M

Neg: -0.33

Q.8	Let A be an invertible 5×5 real matrix. Let B be the matrix obtained by interchanging the second and third columns of A and then adding 3 times the third column to the fifth column. Which ONE of the following statements is TRUE?
(A)	B^{-1} is obtained by interchanging the second and third rows of A^{-1} , and then adding 3 times the third row to the fifth row.
(B)	B^{-1} is obtained by interchanging the second and third rows of A^{-1} , and then adding -3 times the fifth row to the third row.
(C)	B^{-1} is obtained by adding -3 times the third row to the fifth row, and then interchanging the second and third rows of A^{-1} .
(D)	B^{-1} is obtained by adding 3 times the fifth row to the third row, and then interchanging the second and third rows of A^{-1} .

Question 11 (JAM 2025 • Q22 • ID: JAM_2025_Q22)

MCQ

+2 M

Neg: -0.67

Q.22

Let $M = (m_{ij})$ be a 3×3 real, invertible matrix and $\sigma \in S_3$ be the permutation defined by $\sigma(1) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 1$. The matrix $M_\sigma = (n_{ij})$ is defined by $n_{ij} = m_{i\sigma(j)}$ for all $i, j \in \{1, 2, 3\}$.

Then, which one of the following is TRUE?

- (A) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0
- (B) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1
- (C) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1
- (D) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0

Question 12 (JAM 2025 • Q16 • ID: JAM_2025_Q16)

MCQ

+2 M

Neg: -0.67

Q.16

Let $M = \begin{pmatrix} -2 & 0 & 0 \\ 3 & 2 & 3 \\ 4 & -1 & x \end{pmatrix}$ for some real number x . Suppose that -2 and 3 are eigenvalues of M . If $M^3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 125 \\ 125 \end{pmatrix}$, then which one of the following is TRUE?

- (A) $x = 5$, and the matrix $M^2 + M$ is invertible
- (B) $x \neq 5$, and the matrix $M^2 + M$ is invertible
- (C) $x = 5$, and the matrix $M^2 + M$ is NOT invertible
- (D) $x \neq 5$, and the matrix $M^2 + M$ is NOT invertible

Question 13 (JAM 2025 • Q15 • ID: JAM_2025_Q15)

MCQ

+2 M

Neg: -0.67

Q.15

Let $M = \begin{pmatrix} 6 & 2 & -6 & 8 \\ 5 & 3 & -9 & 8 \\ 3 & 1 & -2 & 4 \end{pmatrix}$. Consider the system S of linear equations given by

$$\begin{aligned} 6x_1 + 2x_2 - 6x_3 + 8x_4 &= 8 \\ 5x_1 + 3x_2 - 9x_3 + 8x_4 &= 16 \\ 3x_1 + x_2 - 2x_3 + 4x_4 &= 32 \end{aligned}$$

where x_1, x_2, x_3, x_4 are unknowns.

Then, which one of the following is TRUE?

- (A) The rank of M is 3, and the system S has a solution
- (B) The rank of M is 3, and the system S does NOT have a solution
- (C) The rank of M is 2, and the system S has a solution
- (D) The rank of M is 2, and the system S does NOT have a solution

Question 14 (JAM 2025 • Q13 • ID: JAM_2025_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let $X = \{x \in S_4 : x^3 = id\}$ and $Y = \{x \in S_4 : x^2 \neq id\}$.

If m and n denote the number of elements in X and Y , respectively, then which one of the following is TRUE?

- (A) m is even and n is even
- (B) m is odd and n is even
- (C) m is even and n is odd
- (D) m is odd and n is odd

Question 15 (JAM 2025 • Q7 • ID: JAM_2025_Q7)

MCQ

+1 M

Neg: -0.33

Q.7

Let $M = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & x \end{pmatrix}$ for some real number x .

If 0 is an eigenvalue of M , then $(M^4 + M) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is equal to

(A) $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$

(D) $\begin{pmatrix} 17 \\ 0 \\ 17 \end{pmatrix}$

Question 16 (JAM 2024 • Q50 • ID: JAM_2024_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50

Let

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}.$$

If $p(x)$ is the characteristic polynomial of M , then $p(2) - 1$ equals _____

Question 17 (JAM 2024 • Q44 • ID: JAM_2024_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Consider the 4×4 matrix

$$M = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}.$$

If $a_{i,j}$ denotes the $(i,j)^{\text{th}}$ entry of M^{-1} , then $a_{4,1}$ equals _____ (rounded off to two decimal places).

Question 18 (JAM 2024 • Q37 • ID: JAM_2024_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 For a matrix M , let $\text{Rowspace}(M)$ denote the linear span of the rows of M and $\text{Colspace}(M)$ denote the linear span of the columns of M . Which of the following hold(s) for all $A, B, C \in M_{10}(\mathbb{R})$ satisfying $A = BC$?

- (A) $\text{Rowspace}(A) \subseteq \text{Rowspace}(B)$
- (B) $\text{Rowspace}(A) \subseteq \text{Rowspace}(C)$
- (C) $\text{Colspace}(A) \subseteq \text{Colspace}(B)$
- (D) $\text{Colspace}(A) \subseteq \text{Colspace}(C)$

Question 19 (JAM 2024 • Q16 • ID: JAM_2024_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 Let A be a 6×5 matrix with entries in $\mathbb{R}$ and B be a 5×4 matrix with entries in $\mathbb{R}$. Consider the following two statements.

P: For all such nonzero matrices A and B , there is a nonzero matrix Z such that AZB is the 6×4 zero matrix.

Q: For all such nonzero matrices A and B , there is a nonzero matrix Y such that BYA is the 5×5 zero matrix.

Which one of the following holds?

- (A) Both P and Q are true
- (B) P is true but Q is false
- (C) P is false but Q is true
- (D) Both P and Q are false

Question 20 (JAM 2024 • Q15 • ID: JAM_2024_Q15)

MCQ

+2 M

Neg: -0.67

Q.15

Let $a = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \\ 0 \end{bmatrix}$. Consider the following two statements.

P: The matrix $I_4 - aa^T$ is invertible.

Q: The matrix $I_4 - 2aa^T$ is invertible.

Then, which one of the following holds?

- (A) P is false but Q is true
- (B) P is true but Q is false
- (C) Both P and Q are true
- (D) Both P and Q are false

Question 21 (JAM 2024 • Q9 • ID: JAM_2024_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 Consider the following statements.

P: If a system of linear equations $Ax = b$ has a unique solution, where A is an $m \times n$ matrix and b is an $m \times 1$ matrix, then $m = n$.

Q: For a subspace W of a nonzero vector space V , whenever $u \in V \setminus W$ and $v \in V \setminus W$, then $u + v \in V \setminus W$.

Which one of the following holds?

- (A) Both P and Q are true
- (B) P is true but Q is false
- (C) P is false but Q is true
- (D) Both P and Q are false

Question 22 (JAM 2023 • Q58 • ID: JAM_2023_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 The maximum number of linearly independent eigenvectors of the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is equal to _____.

Question 23 (JAM 2023 • Q56 • ID: JAM_2023_Q56)

NAT

+2 M

Neg: Nil (0)

Q. 56 Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{pmatrix}$$

and B be a 5×5 real matrix such that AB is the zero matrix. Then the maximum possible rank of B is equal to _____.

Question 24 (JAM 2023 • Q14 • ID: JAM_2023_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 The system of linear equations in x_1, x_2, x_3

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{R}$, has

- (A) at least one solution for any α and β
- (B) a unique solution for any β when $\alpha \neq 1$
- (C) NO solution for any α when $\beta \neq 5$
- (D) infinitely many solutions for any α when $\beta = 5$

Question 25 (JAM 2023 • Q13 • ID: JAM_2023_Q13)

MCQ

+2 M

Neg: -0.67

Q. 13 Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -2 & 2 & 2 \end{pmatrix}$$

and $B = A^5 + A^4 + I_3$. Which of the following is NOT an eigenvalue of B ?

- (A) 1
(B) 2
(C) 49
(D) 3

Question 26 (JAM 2023 • Q2 • ID: JAM_2023_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $\mathbf{v}_1, \dots, \mathbf{v}_9$ be the column vectors of a non-zero 9×9 real matrix A . Let $a_1, \dots, a_9 \in \mathbb{R}$, not all zero, be such that $\sum_{i=1}^9 a_i \mathbf{v}_i = \mathbf{0}$. Then the system $A\mathbf{x} = \sum_{i=1}^9 \mathbf{v}_i$ has

- (A) no solution
(B) a unique solution
(C) more than one but only finitely many solutions
(D) infinitely many solutions

Question 27 (JAM 2022 • Q57 • ID: JAM_2022_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Consider the 4×4 matrix $M = \begin{pmatrix} 11 & 10 & 10 & 10 \\ 10 & 11 & 10 & 10 \\ 10 & 10 & 11 & 10 \\ 10 & 10 & 10 & 11 \end{pmatrix}$. Then the value of the determinant of M is equal to _____.

Question 28 (JAM 2022 • Q53 • ID: JAM_2022_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$ and let A^T denote the transpose of A . Let $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be column vectors with entries in $\mathbb{R}$ such that $u_1^2 + u_2^2 = 1$ and $v_1^2 + v_2^2 + v_3^2 = 1$. Suppose

$$Au = \sqrt{2} v \quad \text{and} \quad A^T v = \sqrt{2} u.$$

Then $|u_1 + 2\sqrt{2} v_1|$ is equal to _____. (Rounded off to two decimal places)

Question 29 (JAM 2022 • Q30 • ID: JAM_2022_Q30)

MCQ

+2 M

Neg: -0.67

Q.30

Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0, \lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors of the matrix P corresponding to the eigenvalues λ_1, λ_2 and λ_3 , respectively. Then the entry in the first row and the third column of P is

(A) 0

(B) 1

(C) -1

(D) 2

Question 30 (JAM 2022 • Q26 • ID: JAM_2022_Q26)

MCQ

+2 M

Neg: -0.67

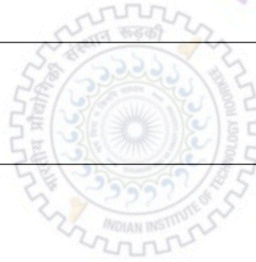
Q.26

For a 4×4 matrix $M \in M_4(\mathbb{C})$, let $\bar{M}$ denote the matrix obtained from M by replacing each entry of M by its complex conjugate. Consider the real vector space

$$H = \{M \in M_4(\mathbb{C}) : M^T = \bar{M}\}$$

where M^T denotes the transpose of M . The dimension of H as a vector space over $\mathbb{R}$ is equal to

- (A) 6
- (B) 16
- (C) 15
- (D) 12



Question 31 (JAM 2022 • Q20 • ID: JAM_2022_Q20)

MCQ

+2 M

Neg: -0.67

Q.20	For $P \in M_5(\mathbb{R})$ and $i, j \in \{1, 2, \dots, 5\}$, let p_{ij} denote the $(i, j)^{\text{th}}$ entry of P. Let $S = \{P \in M_5(\mathbb{R}) : p_{ij} = p_{rs} \text{ for } i, j, r, s \in \{1, 2, \dots, 5\} \text{ with } i + r = j + s\}.$ Then which one of the following is FALSE?
(A)	S is a subspace of the vector space over $\mathbb{R}$ of all 5×5 symmetric matrices.
(B)	The dimension of S over $\mathbb{R}$ is 5.
(C)	The dimension of S over $\mathbb{R}$ is 11.
(D)	If $P \in S$ and all the entries of P are integers, then 5 divides the sum of all the diagonal entries of P .

Question 32 (JAM 2022 • Q18 • ID: JAM_2022_Q18)

MCQ

+2 M

Neg: -0.67

Q.18

Consider the system of linear equations

$$\begin{aligned}x + y + t &= 4, \\ 2x - 4t &= 7, \\ x + y + z &= 5, \\ x - 3y - z - 10t &= \lambda,\end{aligned}$$

where x, y, z, t are variables and λ is a constant. Then which one of the following is true?

- (A) If $\lambda = 1$, then the system has a unique solution.
- (B) If $\lambda = 2$, then the system has infinitely many solutions.
- (C) If $\lambda = 1$, then the system has infinitely many solutions.
- (D) If $\lambda = 2$, then the system has a unique solution.

Question 33 (JAM 2022 • Q17 • ID: JAM_2022_Q17)

MCQ

+2 M

Neg: -0.67

Q.17	For $X, Y \in M_2(\mathbb{R})$, define $(X, Y) = XY - YX$. Let $\mathbf{0} \in M_2(\mathbb{R})$ denote the zero matrix. Consider the two statements: $P : (X, (Y, Z)) + (Y, (Z, X)) + (Z, (X, Y)) = \mathbf{0}$ for all $X, Y, Z \in M_2(\mathbb{R})$. $Q : (X, (Y, Z)) = ((X, Y), Z)$ for all $X, Y, Z \in M_2(\mathbb{R})$. Then which one of the following is correct?
(A)	Both P and Q are true.
(B)	P is true, but Q is false.
(C)	P is false, but Q is true.
(D)	Both P and Q are false.

Question 34 (JAM 2022 • Q16 • ID: JAM_2022_Q16)

MCQ

+2 M

Neg: -0.67

Q.16	Let $P \in M_4(\mathbb{R})$ be such that P^4 is the zero matrix, but P^3 is a nonzero matrix. Then which one of the following is FALSE?
(A)	For every nonzero vector $v \in \mathbb{R}^4$, the subset $\{v, Pv, P^2v, P^3v\}$ of the real vector space $\mathbb{R}^4$ is linearly independent.
(B)	The rank of P^k is $4 - k$ for every $k \in \{1, 2, 3, 4\}$.
(C)	0 is an eigenvalue of P .
(D)	If $Q \in M_4(\mathbb{R})$ is such that Q^4 is the zero matrix, but Q^3 is a nonzero matrix, then there exists a nonsingular matrix $S \in M_4(\mathbb{R})$ such that $S^{-1}QS = P$.

Question 35 (JAM 2022 • Q2 • ID: JAM_2022_Q2)

MCQ

+1 M

Neg: -0.33

Q.2	The rank of the 4×6 matrix $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ with entries in $\mathbb{R}$, is
(A)	1
(B)	2
(C)	3
(D)	4

Question 36 (JAM 2022 • Q1 • ID: JAM_2022_Q1)

MCQ

+1 M

Neg: -0.33

Q.1	Consider the 2×2 matrix $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$. If the eighth power of M satisfies $M^8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$, then the value of x is
(A)	21
(B)	22
(C)	34
(D)	35

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Matrices and Systems of Linear Equations (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+2	1.0 to 1.0	13	MCQ	+2	A	25	MCQ	+2	B
2	NAT	+2	6.0 to 6.0	14	MCQ	+2	B	26	MCQ	+1	D
3	NAT	+2	−1.0 to −1.0	15	MCQ	+1	B	27	NAT	+2	41 to 41
4	NAT	+1	16.0 to 16.0	16	NAT	+1	31 to 31	28	NAT	+2	3.00 to 3.00
5	NAT	+1	3.0 to 3.0	17	NAT	+1	0.15 to 0.18	29	MCQ	+2	C
6	MSQ	+2	A;B;C	18	MSQ	+2	B;C	30	MCQ	+2	B
7	MSQ	+2	A;B;D	19	MCQ	+2	A	31	MCQ	+2	C
8	MCQ	+2	B	20	MCQ	+2	A	32	MCQ	+2	C
9	MCQ	+2	D	21	MCQ	+1	D	33	MCQ	+2	B
10	MCQ	+1	B	22	NAT	+2	3 to 3	34	MCQ	+2	A
11	MCQ	+2	C	23	NAT	+2	2 to 2	35	MCQ	+1	D
12	MCQ	+2	B	24	MCQ	+2	B	36	MCQ	+1	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.